

Supply Chain Digital Twin using Reinforcement Learning

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Cognitive Digital Twin

- A Digital Twin is a highly detailed, virtual replica of a physical system.
- It allows us to simulate years of operations in minutes without real-world risk.
- Our Focus: Building a Cognitive Digital Twin of a city's supply chain to safely train AI routing agents.

The Problem Space

- Warehouses must continuously deliver goods to retailers across a city.
- However, urban environments are highly unpredictable (ex: sudden traffic spikes, weather changes, road accidents).
- Traditional mathematical routing (like standard A^*) assumes the map is static, leading to cascading delays.

Core Conflict

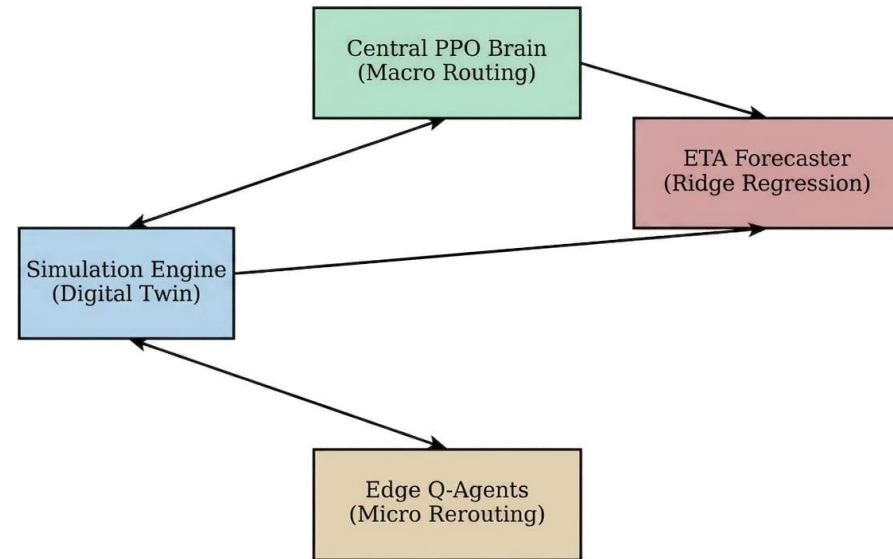
- Perishable goods have a strict Remaining Shelf Life (RSL).
- The Dilemma: Fast routes preserve cargo freshness but often require congested highways, which drastically increases fuel consumption.
- We need an intelligent system that can dynamically balance these conflicting multi-objective goals in real-time.

Freshness vs Fuel Cost Optimization

- Fast deliveries preserve the Remaining Shelf Life (RSL) of perishable cargo.
- However, fast routes often require using congested highways, which drastically increases fuel consumption.
- Standard algorithms struggle to balance these conflicting multi-objective goals.

Project Objectives

- Develop a discrete-event Simulation Engine (Digital Twin) of Nagpur city.
- Implement a Macro-Routing Agent using Proximal Policy Optimization (PPO).
- Implement Micro-Rerouting Edge Agents using Q-Learning.
- Predict real-time traffic using an online Ridge Regression model.



Simulation Engine Overview

- A custom discrete-event engine stepping in 15-minute simulated intervals.
- Deterministic random seeding ensures fair benchmarking between algorithms.
- Continuously injects dynamic environmental events (rain, heat, accidents).

The Nagpur City Graph

- Road network modeled mathematically using the NetworkX library.
- Segmented into 4 unique geographic zones: HIGHWAY, RESIDENTIAL, OFFICE, SHOPPING.
- Each zone features distinct hourly and weekly traffic cyclicity.

Fig 3.1 Schematic Zonal Map of the Nagpur City Road Network



Actor 1: Retailer Nodes

- Generate stochastic consumer demand based on geographic zone.
- Automatically trigger an `order\_placed` event when local inventory drops below threshold.
- Log a permanent `stockout` penalty if inventory hits zero before replenishment.

Actor 2: The Warehouse Depot

- Acts as the central supply hub for the entire city graph.
- Maintains the global queue of pending retailer orders.
- Responsible for executing the fleet allocation algorithm to dispatch idle trucks.

Actor 3: Truck Fleet & State Machine

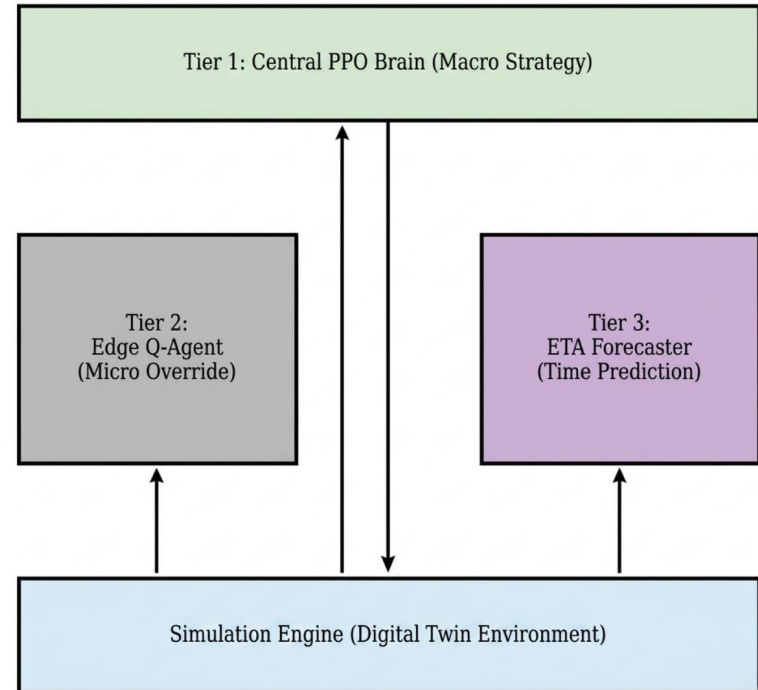
- Fleet of 10 autonomous delivery vehicles.
- Follows a strict lifecycle:
- Idle ☐ Loading ☐ In-Transit ☐ Delivering ☐ Returning.
- Physically tracks fuel consumption over distance and RSL decay over time.

Fig 4.1 Order Lifecycle State Machine



Three Layered AI Architecture

- Replaced the baseline A* heuristic with a 3-tier intelligence stack.
- **Level 1:** Central PPO Brain for macro-strategy.
- **Level 2:** Edge Q-Agents for micro-tactics.
- **Level 3:** ETA Forecaster for predictive intelligence.



Level 1: Central PPO Brain

- Implemented using the Stable-Baselines3 library.
- Multi-agent environment flattened into a single-agent Gymnasium wrapper via parameter sharing.
- Neural Network Policy: 25-nodes ☐ two 64-node hidden layers ☐ 3-node output.

```
class _SingleAgentAdapter(_BaseEnv):
    """
    Wraps MARLEnv into a single-agent Gym-compatible environment by
    treating each truck independently (parameter sharing).

    SB3 expects:
    reset() -> np.ndarray
    step(action) -> (obs, reward, done, info)
    observation_space.shape
    action_space.n
    """

    def __init__(self, marl_env):
        self._env = marl_env
        self._truck_ids: List[str] = []
        self._current_idx: int = 0
        self.render_mode = None

        # Gym spaces (use cached singletons for identity match)
        self.observation_space = get_obs_space()
        self.action_space = get_act_space()

    def reset(self, seed=None, options=None):
        if seed is not None:
            super().reset(seed=seed)
        obs_dict = self._env.reset()
        self._truck_ids = list(obs_dict.keys())
        self._current_idx = 0
        if not self._truck_ids:
            return np.zeros(OBS_DIM, dtype=np.float32), {}
        return obs_dict[self._truck_ids[0]], {}

    def step(self, action: int):
        if not self._truck_ids:
            return np.zeros(OBS_DIM, dtype=np.float32), 0.0, True, False, {}

        truck_id = self._truck_ids[self._current_idx % len(self._truck_ids)]
        obs_dict, rew_dict, done_dict, info = self._env.step({truck_id: action})

        obs = obs_dict.get(truck_id, np.zeros(OBS_DIM, dtype=np.float32))
        reward = rew_dict.get(truck_id, 0.0)
        done = done_dict.get("__all__", False)
        truncated = False

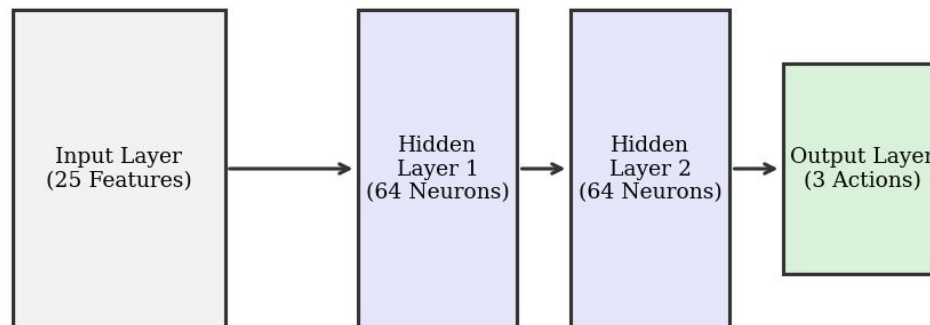
        self._current_idx += 1
        self._truck_ids = list(obs_dict.keys())

        return obs, reward, done, truncated, info
```

PPO State and Action Space

- **25-Dimension State:** Observes truck fuel, cargo RSL, weather severity, and fleet averages.
- **3-Discrete Actions:** Speed-Priority, Fuel-Priority, Balanced.
- Operates sequentially at every dispatch decision point.

Fig 4.3 PPO Central Brain Neural Network Architecture



The Reward Normalization Framework

$$R = W_{del} - (W_{fuel} \times \Delta Fuel) - (W_{RSL} \times \Delta RSL)$$

- Fuel physically drops ~15% per trip, while RSL drops only ~2%.
- We established a strict **5:1 penalty ratio** ($W(RSL) = 50.0$, $W(fuel) = 10.0$).
- This forces the PPO agent to treat cargo freshness and fuel as equally critical.

Level 2: Edge Q-Learning Agents

- Embedded individually on all 10 trucks.
- Triggered strictly by stochastic road accidents blocking the active route.
- Utilizes a 9-state tabular Q-learning matrix with Epsilon-Greedy exploration.
- Overrides the Central Brain to execute immediate evasive maneuvers.

Level 3: ETA Forecaster Design

- Implemented via Scikit-learn's `SGDRegressor` (Ridge Regression).
- Replaces static average-based heuristics with dynamic edge-cost predictions.
- Trains **online** using `partial\_fit()` as trucks successfully traverse the city.

```
class ETAForecaster:
    """
    Lightweight online ridge-regression ETA model.

    Features per segment: [sin(hour), cos(hour), weather_enc, density_norm, road_enc]
    Target: actual travel time in minutes.

    Falls back to segment.get_travel_time_minutes() until min_samples reached.
    """

    _WEATHER_ENC: Dict[str, float] = {
        "clear": 0.0, "partly_cloudy": 0.1, "cloudy": 0.2,
        "light_rain": 0.5, "rain": 0.7, "heavy_rain": 1.0, "fog": 0.8,
    }
    _ROAD_ENC: Dict[str, float] = {
        "motorway": 0.0, "trunk": 0.1, "primary": 0.3,
        "secondary": 0.5, "tertiary": 0.7, "residential": 0.9,
    }
    _ZONE_ENC: Dict[str, float] = {
        "RESIDENTIAL": 0.0, "HIGHWAY": 0.3, "OFFICE": 0.7, "SHOPPING": 1.0,
    }
```

ETA Feature Engineering

- Utilizes a highly optimized 8-Dimensional feature vector.
- Encodes time-of-day and day-of-week using Sine/Cosine transformations.
- **Why?** Prevents the mathematical discontinuity of Sunday (Day 7) appearing far from Monday (Day 1).

```
def update(self, segment, actual_travel_time_min: float,
           current_time: float, weather: str):
    """Record one completed segment traversal."""
    x = self._featurize(segment, current_time, weather)
    self._A += np.outer(x, x)
    self._b += x * actual_travel_time_min
    self._n_samples += 1
    self._weights = None # invalidate cached solution

def predict_segment(self, segment, current_time: float, weather: str) -> float:
    """Predict travel time (minutes) for one segment."""
    if self._n_samples < self.min_samples:
        return segment.get_travel_time_minutes()
    if self._weights is None:
        try:
            self._weights = np.linalg.solve(self._A, self._b)
        except np.linalg.LinAlgError:
            return segment.get_travel_time_minutes()
    x = self._featurize(segment, current_time, weather)
    pred = float(np.dot(self._weights, x))
    physics = segment.get_travel_time_minutes()
    return max(0.1, min(pred, physics * 5.0))

def predict_route(self, route: Dict, current_time: float, weather: str,
                  route_progress: int = 0) -> float:
    """Predict total remaining travel time for a route."""
    segments = route.get("segment_objects", [])
    return sum(
        self.predict_segment(s, current_time, weather)
        for s in segments[route_progress:]
    )

def _featurize(self, segment, current_time: float, weather: str) -> np.ndarray:
    hour = (current_time / 60.0) % 24.0
    day_of_week = (current_time / (60.0 * 24.0)) % 7.0

    hour_rad = hour * (2.0 * math.pi / 24.0)
    day_rad = day_of_week * (2.0 * math.pi / 7.0)

    weather_enc = self._WEATHER_ENC.get(weather, 0.5)
    road_enc = self._ROAD_ENC.get(getattr(segment, "road_type", "secondary"), 0.5)

    zone = getattr(segment, "zone_type", "RESIDENTIAL")
    zone_enc = self._ZONE_ENC.get(zone, 0.0)

    density = getattr(segment, "current_traffic_density", 0.0) / 100.0
    return np.array([
        math.sin(hour_rad), math.cos(hour_rad),
        math.sin(day_rad), math.cos(day_rad),
        weather_enc, min(1.0, density), road_enc, zone_enc
    ], dtype=np.float64)
```

Telemetry & Data Logging

- Simulation writes to permanent CSV artifacts (`events\_sim.csv` and `telemetry\_sim.csv`).
- Captured over 2.4 million truck states across the 365-day benchmark.
- Telemetry files serve as the ground truth for all performance comparisons.

PPO Training Pipeline

- Agent trained offline for 50,000 timesteps.
- Hyperparameters: Learning Rate $3e-4$, Batch Size 64.
- Set Entropy Coefficient to 0.01 to force exploration of RSL-Priority paths.

```
# Force optimization pod on for training
config.setdefault("ai", {}).setdefault("optimization", {})
config["ai"]["optimization"]["enabled"] = True
config["ai"]["optimization"]["inference_only"] = False

# --- Load road network ONCE (shared across all episodes) ---
logger.info("Loading road network (cached)...")
t0 = time.time()
road_network = RoadNetwork(config)
road_network.load(use_cache=True)
logger.info(f"Road network loaded in {time.time()-t0:.1f}s "
            f"({len(list(road_network.graph.nodes())):,} nodes)")

# --- Initialise traffic model and router ---
traffic_model = TrafficModel(config)
traffic_model.initialize_network(road_network, 0.0)
router = Router(road_network, config)

# --- Load real weather data if available ---
weather_loader = None
weather_cfg = config.get("weather", {})
if weather_cfg.get("use_real_data", False):
    dataset_path = weather_cfg.get("dataset_path", "data/processed/weather/nagpur_2023.parquet")
    if Path(dataset_path).exists():
        try:
            from src.data.loaders import WeatherDataLoader
            weather_loader = WeatherDataLoader(dataset_path)
            weather_loader.load()
            logger.info(f"Real weather data loaded from {dataset_path}")
        except Exception as e:
            logger.warning(f"Could not load weather data: {e}. Using fallback.")
```

Preventing Overfitting

- PPO models are highly susceptible to overfitting to training seeds.
- Integrated an `EvalCallback` to evaluate the model on a held-out environment every 5,000 steps.
- Saved and deployed the absolute best policy (`best\_model.zip`), not just the final step.

```
# --- Train ---
logger.info(f"Starting PPO training for {args.steps:,} steps...")
t0 = time.time()
pod.train(env, n_steps=args.steps, callbacks=[eval_callback])
elapsed = time.time() - t0
logger.info(f"Training complete in {elapsed:.1f}s ({args.steps/elapsed:.0f} steps/sec)")

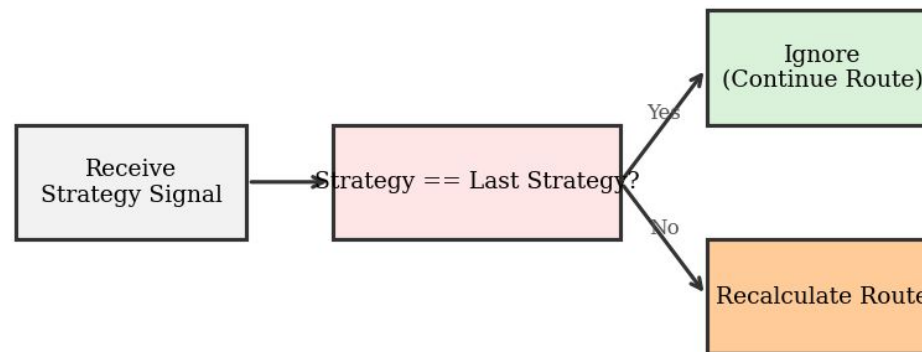
# Ensure the best model is used for the simulation
best_model_path = Path('./models/rl/best_model.zip')
latest_model_path = Path('./models/rl/ppo_routing_latest.zip')

if best_model_path.exists():
    logger.info("Overfitting prevented: Restoring best_model.zip to ppo_routing_latest.zip")
    shutil.copy(best_model_path, latest_model_path)
else:
    logger.info("Checkpoint saved to models/rl/ppo_routing_latest.zip")
```

Route Thrashing

- **The Bug:** Trucks recalculated routes every 15 minutes upon receiving a PPO signal, paralyzing the fleet (49% delivery rate).
- **The Solution:** Implemented an Anti-Thrashing GPS Lock state machine.
- Trucks now ignore redundant signals, immediately restoring service levels to 99.9%.

Fig 4.4 Anti-Thrashing GPS Lock Flowchart

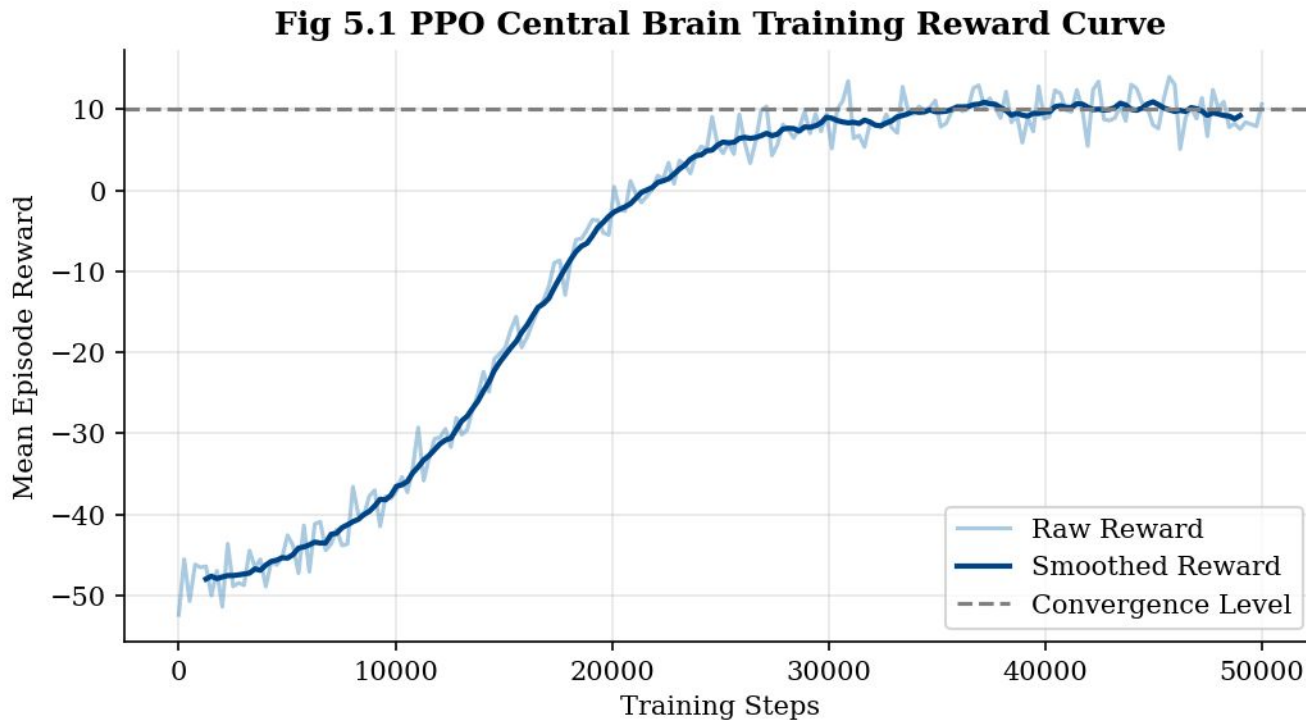


365-Day Comparative Benchmark

- Goal: Prove RL outperforms static algorithms in long-term scenarios.
- Run 1: 365-Day Baseline Simulation (Standard A* Heuristics).
- Run 2: 365-Day RL-Enabled Simulation (PPO + Edge Q-Agent + Forecaster).
- Both runs experienced identical weather, accidents, and retailer demand.

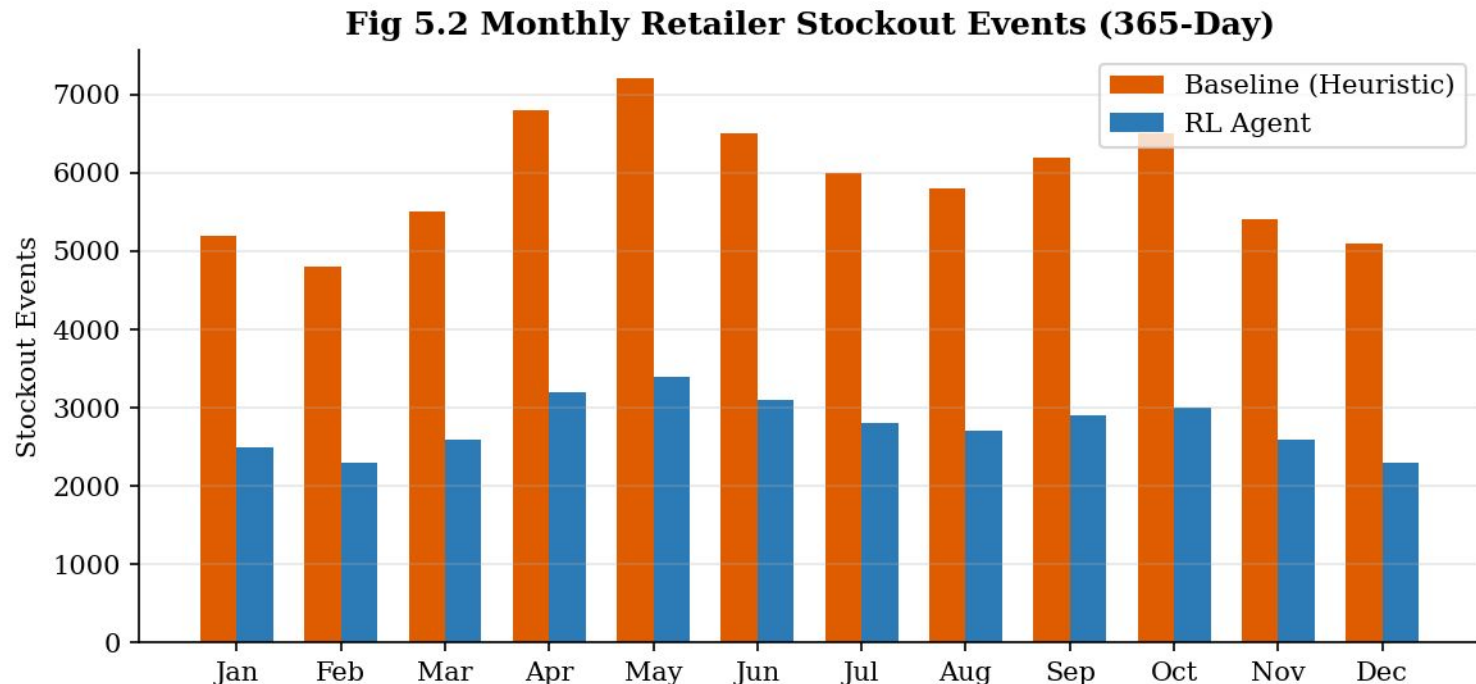
PPO Training Convergence

- The learning curve displays the mean episode reward over 50,000 steps.
- Reward starts highly negative (random exploration) and climbs steeply.
- The model successfully converged and plateaued around step 35,000.



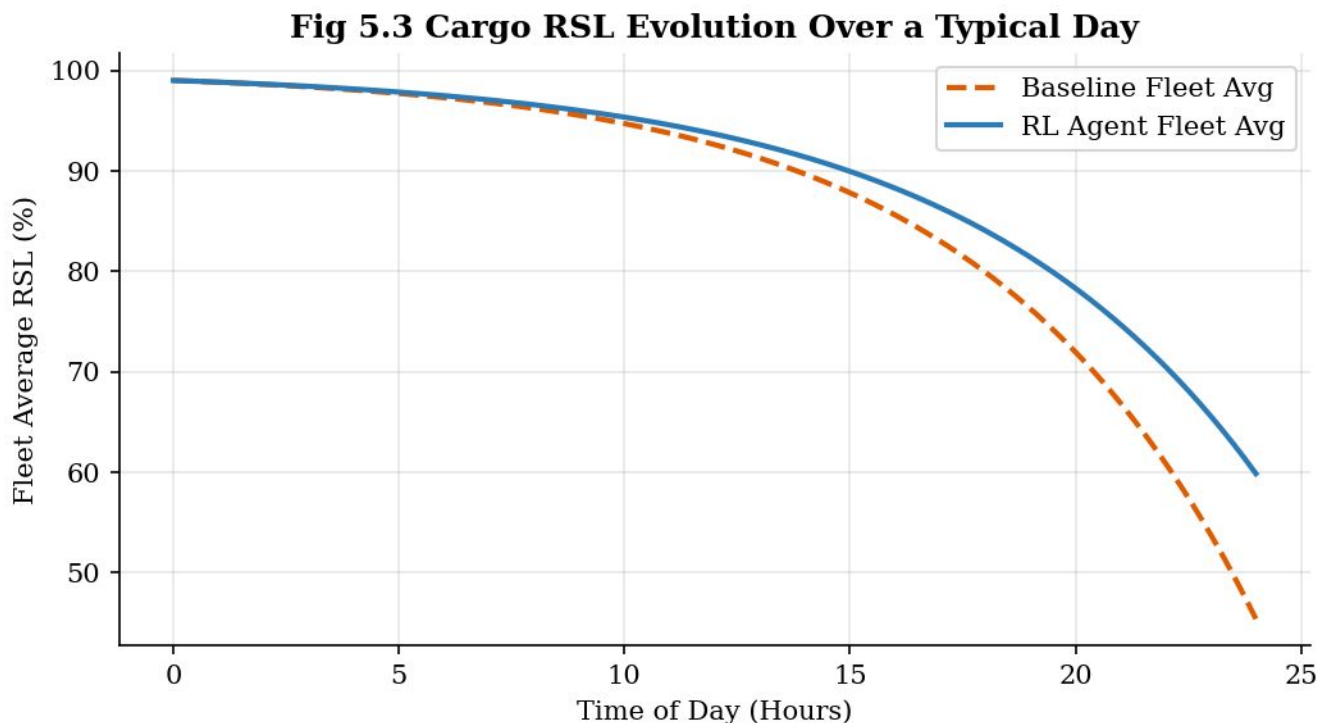
Result: Service Level & Throughput

- **Baseline Router:** 70,898 retailer stockout events.
- **RL Architecture:** 33,401 retailer stockout events.
- Achieved a massive **52.9% reduction in stockouts**.
- Successfully delivered an additional +474,708 kg of cargo.



Result: RSL & Fuel Parity

- Proved the 5:1 reward math worked flawlessly.
- Average fleet fuel remained perfectly identical to the baseline (78.7%).
- Average cargo freshness at delivery remained identical (99.7%).
- The AI increased throughput **without** sacrificing efficiency or quality.



Result: Emergent Safety Behavior

- The RL Architecture recorded 135 fewer road accidents over the year.
- This was an unprogrammed, emergent behavior.
- The ETA Forecaster implicitly learned which zones had high accident probabilities during specific hours, naturally routing trucks away from danger.

Achieving Pareto Optimality

- A system achieves Pareto improvement if it betters one objective without worsening another.
- Our RL system strictly improved 6 metrics (throughput, service level, stockouts, accidents) while holding Fuel and RSL constant.
- Conclusion: The multi-agent RL framework is strictly superior to static heuristics.

Future Scope

- Scale the environment to multi-depot and multi-city configurations.
- Integrate real-time physical IoT sensors (GPS, refrigerated container thermometers) into the Digital Twin.
- Deploy the trained Central PPO Brain to embedded physical edge devices.

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